

Cellular Expert

2. CE Express. Creating and Managing Network Objects

1. Objective

This module explains and demonstrates how to create, edit, duplicate, and manage network objects in CE Express using object templates.

By the end of this exercise, participants will be able to:

- Create and use object templates effectively
- Create sites and cells with correct RF parameters
- Duplicate and reposition network objects efficiently
- Modify object parameters using both forms and tables
- Apply meaningful network symbology for RF analysis

2. Initial Data and Prerequisites

The following prerequisites are assumed:

- A prepared workspace created in: **01. CE Express – Creating Workspace**
- Available geodata (terrain, obstacles and clutter)
- User permissions to create and edit network objects

3. Understanding the Network Objects Concept

In CE Express, **network objects** represent the physical and logical elements of a radio network. They form the foundation for all RF planning, simulation, and optimization activities performed in the workspace.

Network objects typically include:

- **Sites** – physical locations where radio equipment is installed
- **Cells / Sectors** – logical radio entities defining coverage and capacity
- **Antennas** – radiation characteristics, patterns, and beamforming behavior
- **Links and relations** – logical associations between network elements

These objects are tightly coupled with:

- **Geodata** (terrain, clutter, obstacles), which influences propagation behavior

- **Technology definitions** (LTE, 5G NR, TDD/FDD)
- **Templates**, which standardize configuration and reduce engineering errors

Why Templates Matter

Object templates encapsulate best-practice RF configurations, such as:

- Standard power levels and bandwidths
- Technology-specific parameters (e.g. NR numerology, MIMO layers)
- Approved antennas and prediction models

Using templates ensures:

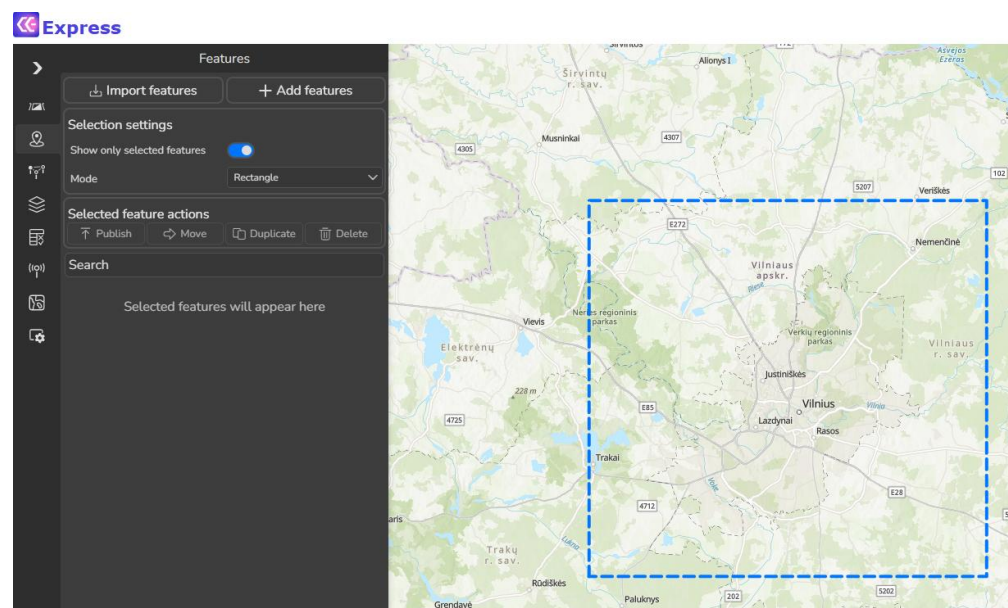
- Consistency across large networks
- Faster rollout and scenario creation
- Reduced risk of misconfiguration during planning and optimization

3.1 Step 1 – Open the Workspace

1. Open the CE Express application:

https://cecom2.cellular-expert.com/ce_express/

2. From the workspace list, select the workspace created in the previous exercise.

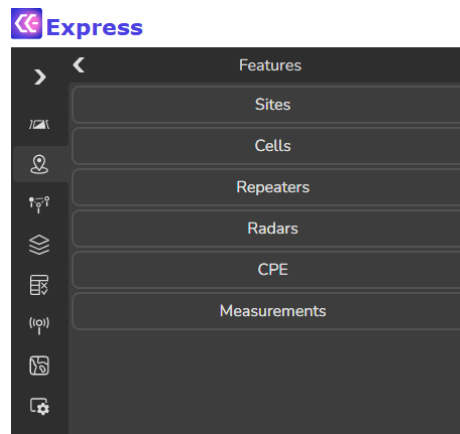


Best practice: Always confirm the active workspace before object creation to avoid placing objects in the wrong project.

3.2 Step 2 – Creating Network Objects

3.2.1 Creating a Site Object

1. Open the Features tool.
2. Click Add Feature and select Sites.

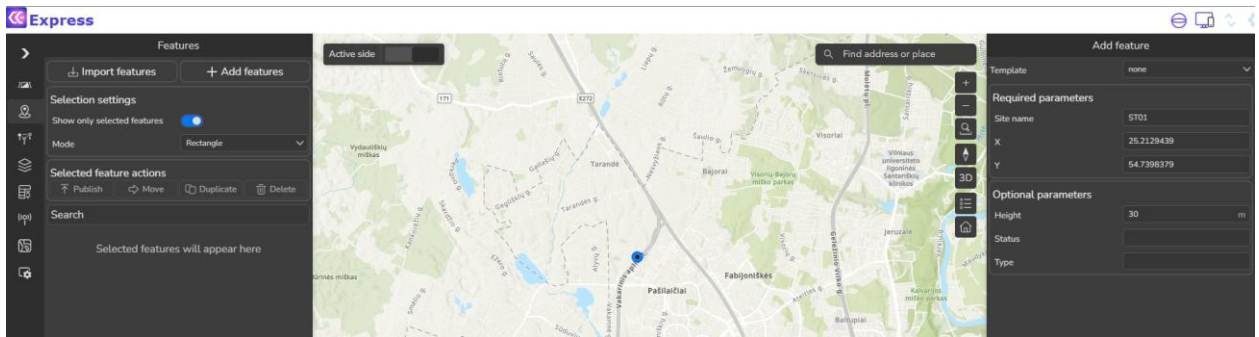


Define the following parameters:

- Site Name: ST01
- X (Longitude): 25.2129439
- Y (Latitude): 54.7398379
- Height (m): 30

A screenshot of the 'Add feature' dialog box. At the top, the title 'Add feature' is centered. Below it, there is a 'Template' dropdown menu set to 'none'. The dialog is divided into two sections: 'Required parameters' and 'Optional parameters'. Under 'Required parameters', there are three input fields: 'Site name' with the value 'ST01', 'X' with the value '25.2129439', and 'Y' with the value '54.7398379'. Under 'Optional parameters', there are three input fields: 'Height' with the value '30' and a unit 'm' dropdown, 'Status', and 'Type'.

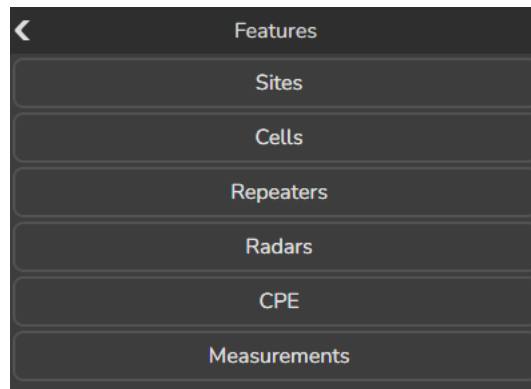
3. Click Accept to create the site.



RF note: Site height represents the average antenna mounting height above ground and directly impacts coverage and LOS conditions.

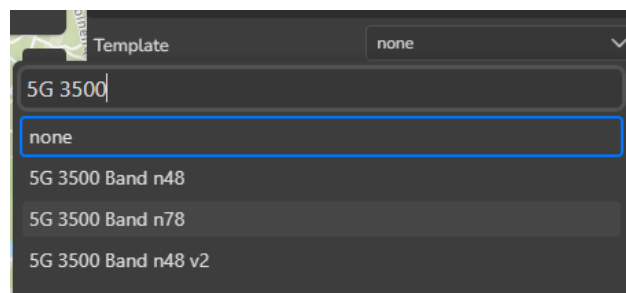
3.2.2 Creating Cell Objects Using Templates

1. In the Features tool, click Add Feature again.



2. Select Cells.
3. In the template selection window, choose:

5G 3500 – Band n78



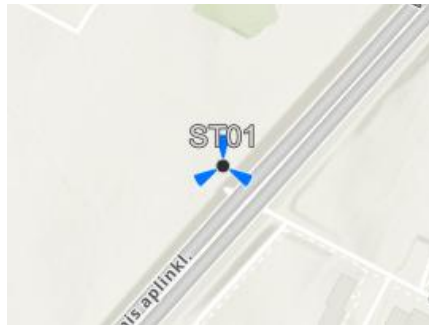
Templates pre-fill technology-specific parameters, ensuring consistency and reducing configuration errors.

3.2.3 Creating a 3-Sector 5G Site

Then create three new cells based on table below (once you enter all parameters for one cell object press Accept button, then change parameters for second cell and press Accept button, and do it for third cell).

Parameter	Value		
Cell name	Cx01	Cx02	Cx03
X	25.2129439	25.2129439	25.2129439
Y	54.7398379	54.7398379	54.7398379
Azimuth, deg	0	120	240
Height, m	30	30	30
Downtilt, deg	1	1	1
El. Downtilt, deg	0	0	0
Frequency, MHz	3500	3500	3500
Power, dBm	40	40	40
Misc Loss, dB	0	0	0
Bandwidth, MHz	100	100	100
Noise Figure, dB	6	6	6
Downlink duplex factor	0,6	0,6	0,6
Subcarrier spacing, kHz	60	60	60
TX MIMO	32	32	32
RX MIMO	32	32	32
Active Antenna Effect	6	6	6
Cell Load, %	30	30	30
Color index			
Technology	5G	5G	5G
Prediction model	CEC ITU-R: 3km radius	CEC ITU-R: 3km radius	CEC ITU-R: 3km radius
Frequency group	3500	3500	3500

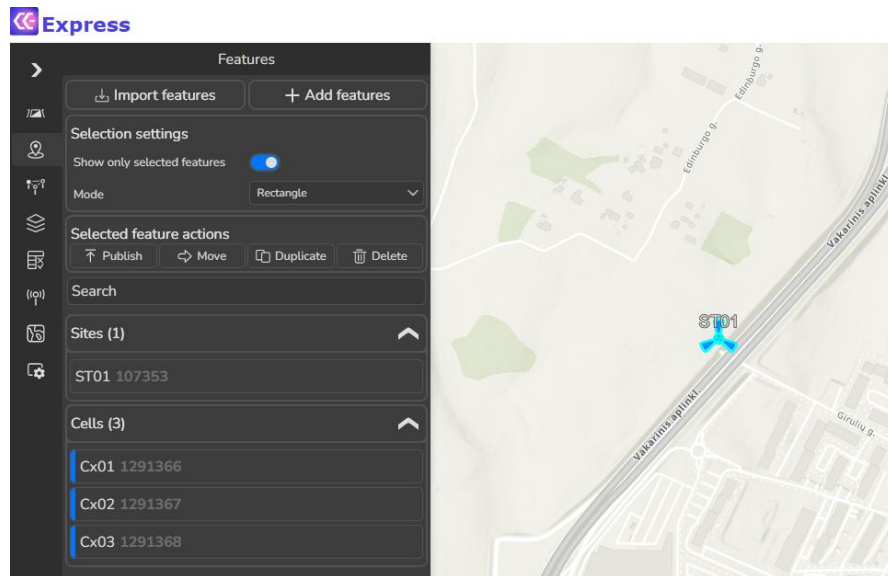
Antenna	<i>S4-90M-R1- V2_BC-3300- 3600- P0_02DT_3500</i>	<i>S4-90M-R1- V2_BC-3300-3600- P0_02DT_3500</i>	<i>S4-90M-R1- V2_BC-3300- 3600- P0_02DT_3500</i>
Carriers	[]	[]	[]
SiteID			
Duplex Mode	TDD	TDD	TDD



4. After creating all three cells, click Cancel to close the Add Feature tool.

3.3 Step 3 – Selecting and Reviewing Objects

1. Select the newly created site and cells on the map.



2. Verify their placement and orientation.

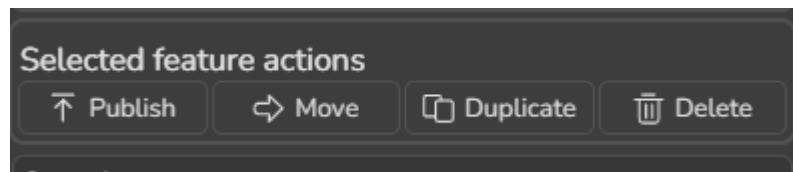
3.4 Step 4 – Duplicating Network Objects

Duplicating objects is commonly used for:

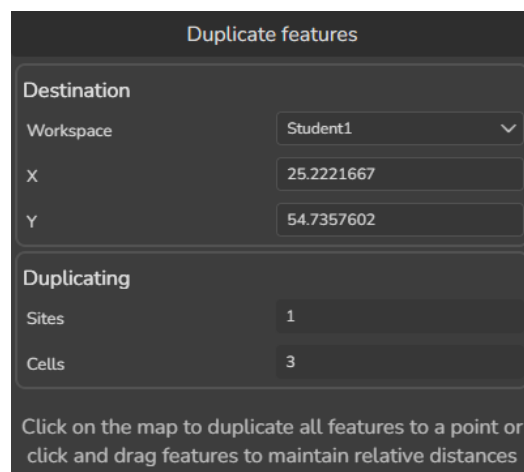
- Multi-site rollout scenarios
- Rapid what-if analysis
- Dense urban planning

Duplicate Procedure

1. Keep all created objects selected.
2. In the Features tool, click Duplicate.

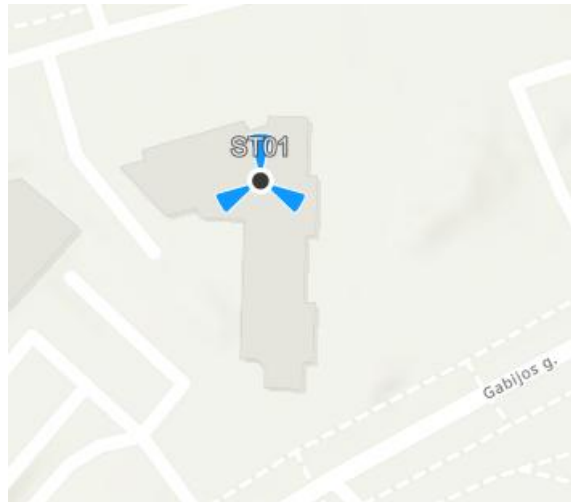


3. In the Duplicate dialog, define:
 - Workspace: current workspace
 - X: 25.2221667
 - Y: 54.7357602

A screenshot of a "Duplicate features" dialog box. It has two main sections: "Destination" and "Duplicating". In the "Destination" section, there is a "Workspace" dropdown menu set to "Student1", and input fields for "X" (25.2221667) and "Y" (54.7357602). In the "Duplicating" section, there are input fields for "Sites" (1) and "Cells" (3). At the bottom, there is a text instruction: "Click on the map to duplicate all features to a point or click and drag features to maintain relative distances".

4. Click Accept.

All selected objects are duplicated at the new location.



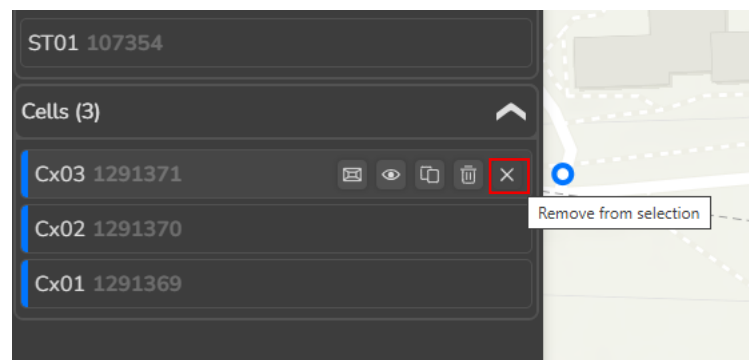
3.5 Step 5 – Moving Individual Objects

Objects can be repositioned individually or as a group.

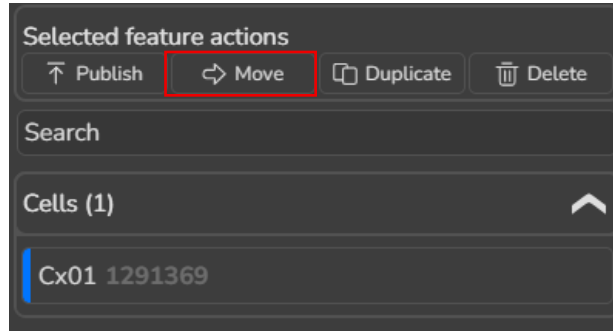
In this exercise, cells will be placed at different corners of a building.

3.5.1 Moving a Single Cell

1. Select duplicated objects.
2. Remove all but Cx01 from the selection.



3. Click the Move tool.



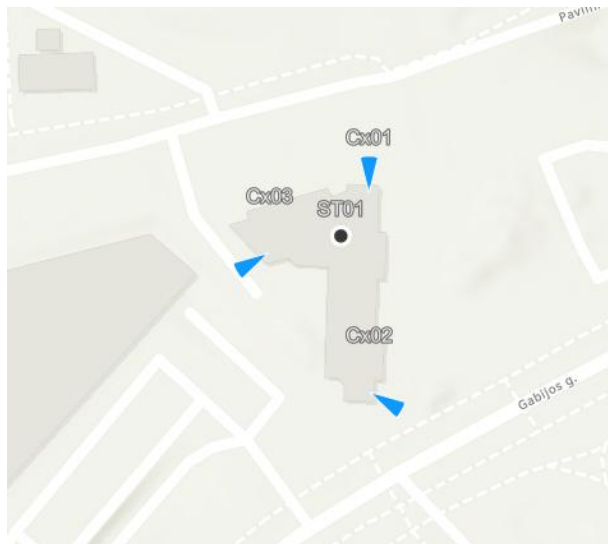
Define new coordinates:

- X: 25.2222847
- Y: 54.7358624

4. Click Accept.

Repeat the process for:

- Cx02 → X: 25.2222847, Y: 54.7353823
- Cx03 → X: 25.2218663, Y: 54.7357199



3.6 Step 6 – Modifying Object Parameters

After objects are created and positioned, the next important task is updating and refining their parameters. Correct parameters ensure that objects are clearly identified, consistently configured, and ready for further analysis, visualization, or reporting across different user roles.

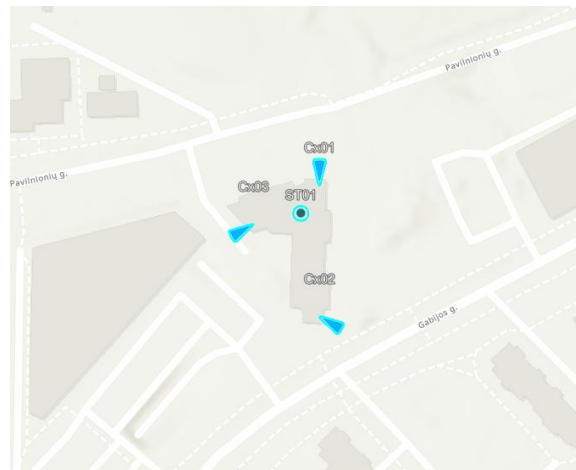
CE Express provides two complementary ways to modify object parameters:

- **Object-level editing** – detailed changes for individual objects
- **Table-based editing** – fast and efficient updates for multiple objects

Both methods are commonly used together in real projects.

3.6.1 Editing via Object Properties

1. Select newly duplicated/moved objects on the map.



2. Click on the duplicated ST01 site in Features tool.
3. In the Features tool, change the site name to: ST02

Edit feature	
Template	none
Required parameters	
Site name	ST02
X	25,2221667
Y	54,7357602
Optional parameters	
Height	30 m
Status	
Type	
Read-only parameters	
ID	107354

4. Click Accept.

Clear and unique object names help all users quickly understand the network structure and avoid confusion.

4. Click on the Cx01 cell in Features tool and modify the following parameters:

- Name: Cv001
- Azimuth: 20
- Frequency: 1800

Edit feature

Template: none

Required parameters

Cell name: Cx001

X: 25,2222847

Y: 54,7358624

Azimuth: 20 deg

Optional parameters

Height: 30 m

Downtilt: 0 deg

EL Downtilt: 0 deg

Frequency: 1800 MHz

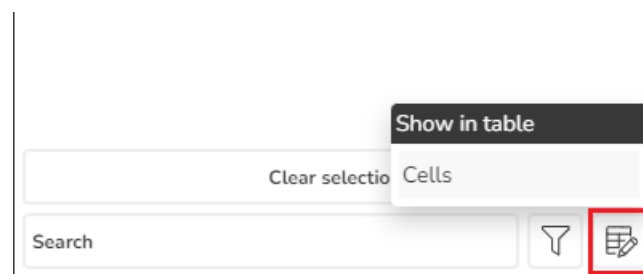
5. Click Accept.

Changing these parameters updates how the object is represented and interpreted in the project.

3.6.2 Editing via Table View

Table view is especially useful when working with multiple objects or large projects.

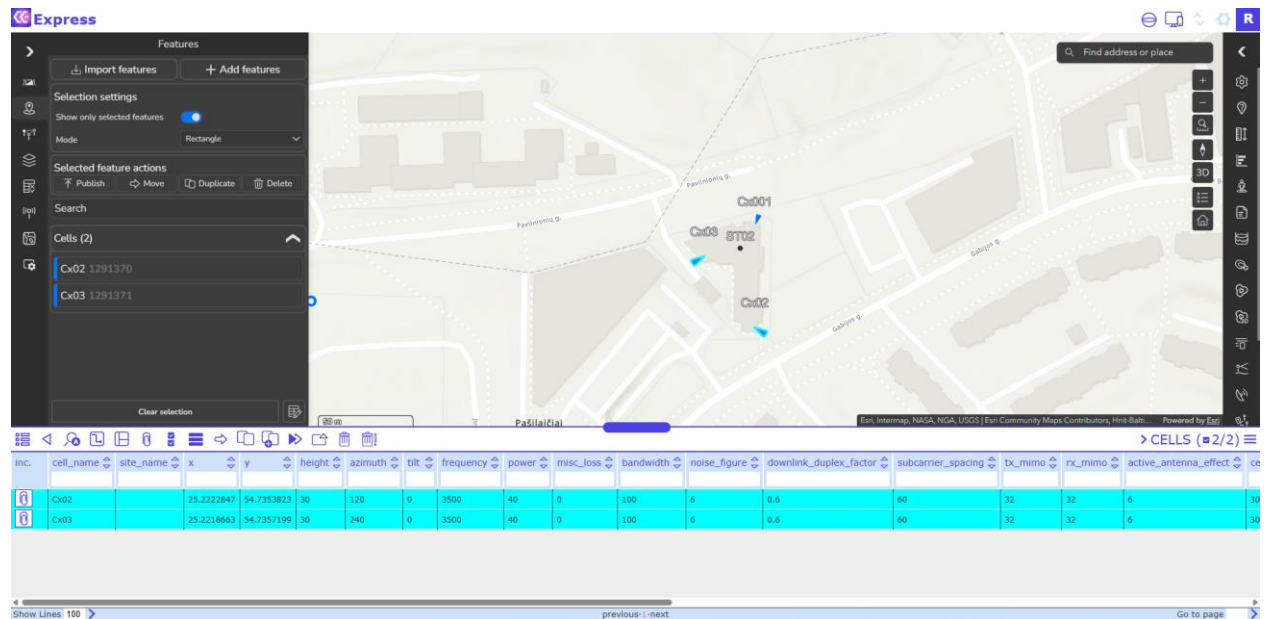
1. In the Features tool, click Show in Table.



2. Select Cells.

The table and the map are synchronized:

- Selecting an object in the table highlights it on the map
- Selecting an object on the map highlights it in the table



Update parameters as follows:

- **Cx02:**
 - Cell Name → Cv002
 - Azimuth → 160
- **Cx03:**
 - Cell Name → Cv003
 - Azimuth → 250

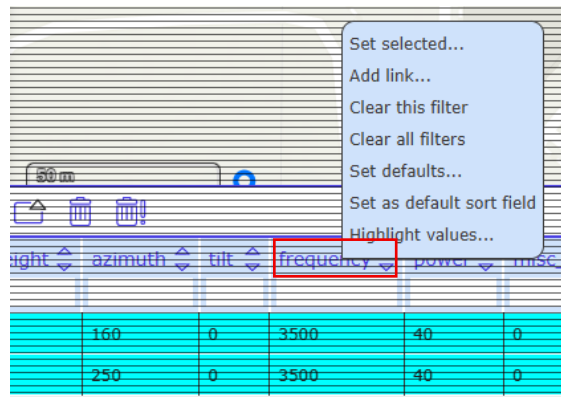
This approach makes it easy to review and adjust values while maintaining full spatial awareness.

3.6.3 Bulk Editing Multiple Objects

1. Select multiple cells in the table (left click on them).

inc.	cell_name	site_name	x	y	height	azimuth	tilt	frequency	power	misc_loss
1	Cx002		25.2222847	54.7353823	30	160	0	3500	40	0
1	Cx003		25.2218663	54.7357199	30	250	0	3500	40	0

2. Left click on the Frequency field.



3. From the context menu, choose Set selected.

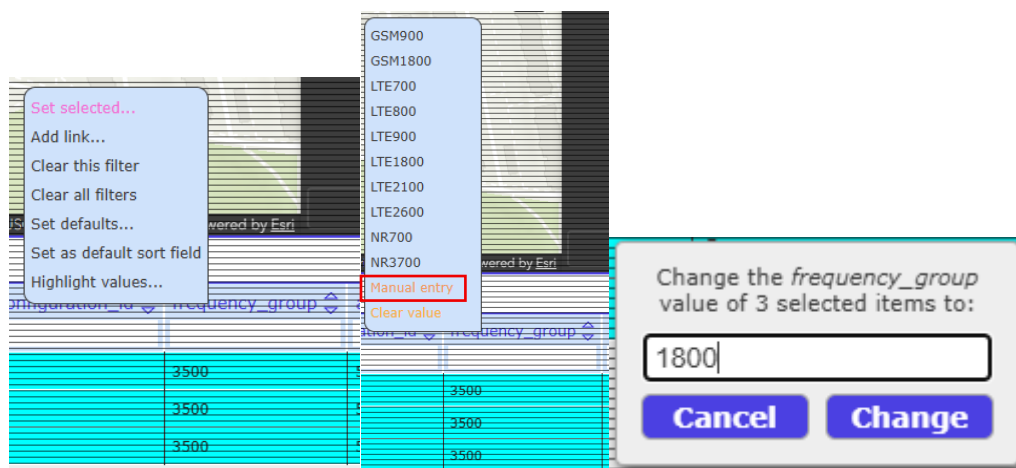
4. Enter the new value: 1800 MHz.

Change the *frequency* value of 2 selected items to:

Cancel **Change**

5. Click Change.

6. Select Cells again and do the same for frequency_group field. Define value 1800.



3.7 Step 7 – Managing Network Symbolology

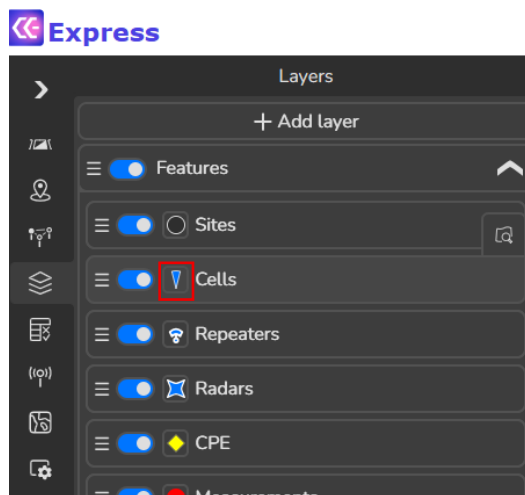
Network symbology defines how objects are visually displayed on the map. While symbology does not change object data, it plays a key role in making complex projects easier to understand for different audiences, including technical teams, decision-makers, and non-technical users.

Well-designed symbology helps to:

- Distinguish between technologies and frequency layers
- Improve map readability in dense areas
- Support presentations, reviews, and decision-making

3.7.1 Configuring Cell Symbology

1. Open the Layers tool.
2. Click on the Cells icon to open the Cell Symbolization panel.

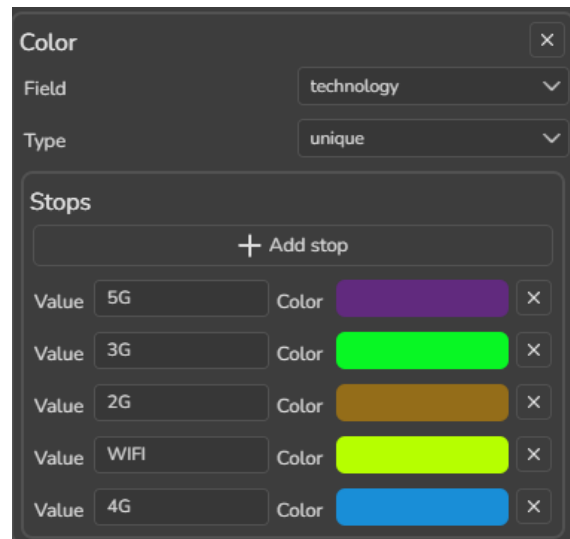


It will open a Cells Symbolization tool that customers can define on how network objects should be visualized.

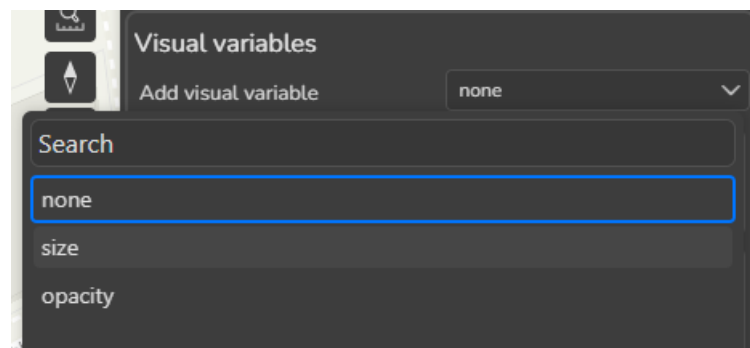
Define:

- Draw order
 - Order by: frequency
 - Order: descending
- Visual variables: Color

- Technology
- Stops: define as in the picture below



- In Visual variables add a new parameter: Size



- Field: frequency
- Stops: define as in the picture below

Size

Field frequency

Stops

+ Add stop

Value	6000	Size	15	
Value	4000	Size	18	
Value	2600	Size	22	
Value	1900	Size	28	
Value	900	Size	33	
Value	700	Size	37	
Value	0	Size	50	

Press Accept button.

Cells symbology will be updated

